Verigate PortalDot

Authorization middleware for AI-agent payments on PortalDot V3

A proposal is not authority.

Verigate evaluates the action. The bridge verifies signed evidence. A dedicated Solidity guard enforces the exact approved transfer.

VERIGATE	BRIDGE	PORTALDOT
Ed25519 decision	Exact action binding	EIP-712 guard

TESTNET

420420777

PortalDot V3 EVM testnet

Contract deployed • Receipt verified

Track fit

Modular Cloud Solutions and Middleware

Built around a narrow, reusable control point between AI intent and token execution.

Preventing unrestricted agent spending

The authorization must bind the decision to the exact thing the executor is allowed to do.

01 RECIPIENT SUBSTITUTION

An approved payment must not be reused with a different destination.

02 AMOUNT TAMPERING

The amount must remain the same from signed intent to contract call.

03 REPLAY OR EXPIRY

A valid authorization should not be usable again or after its deadline.

Design principle

A signed ALLOW is necessary but not enough: its authority must be short-lived, narrowly scoped, and bound to a recipient, amount, chain and contract.

A narrow path from decision to execution

Off-chain verification and on-chain enforcement have separate, explicit trust boundaries.

1

AGENT INTENT

Proposed action

2

VERIGATE

Signed ALLOW + execution authorization

3

BRIDGE

Verify Ed25519; bind exact action

4

EIP-712

Chain, guard, recipient, amount, nonce, deadline

5

GUARD

Reject invalid/replayed or altered transfer

WARN / BLOCK / invalid signature / mismatched action

The bridge refuses to sign; the contract cannot independently override this off-chain decision.

Why PortalDot V3 — and where this project fits

The project uses PortalDot's execution environment; it does not overstate a cross-chain bridge that is not implemented.

- **EVM-compatible contract execution**

  The guard is written in Solidity and uses EIP-712 signatures bound to chain ID and contract address.

- **Substrate-aware operational checks**

  The tooling separately checks the configured Substrate WebSocket endpoint and genesis hash before sensitive network operations.

- **Reusable middleware contribution**

  Other agent-enabled applications can reuse a strict authorization boundary instead of giving an agent a direct, unrestricted transfer path.

TRACK

Modular Cloud Solutions and Middleware

Not claiming:

A deployed multi-chain asset bridge, verified wallet mapping, production readiness, or an independent audit.

PortalDot V3 deployment and transfer evidence

The public testnet now confirms a Verigate-authorized PortalDotAgentGuard call and its TransferExecuted receipt.

NETWORK
PortalDot V3 EVM testnet

CHAIN ID
420420777

CONTRACT
0x3f2f4844...F09655b

DEPLOYMENT TX
0xb1cf9cca...1fcdce3c

RUNTIME CODE HASH
0xb3d71ff8...f7f7671

TRANSFER TX
0xbe3ade2b...9914cc14

TRANSFER RECEIPT
Success • block 130741

READ-BACK

2,153 bytes

Runtime bytecode

authoritySigner()

On-chain value matched the configured signer.

Console
node-console.portaldot.dev

What has been verified — and what has not

Local tests remain separate from the newly confirmed PortalDot public testnet transfer.

156
local tests passed

ALLOW
valid signed artifacts

BLOCK
execution refused

- **Pre-broadcast checks passed**
  The executor re-simulated the exact signed authorization and refreshed the fee budget before sending.

- **Public PortalDot transfer confirmed**
  Receipt status 1 at block 130741; TransferExecuted matched the action ID, recipient and amount.

IMPORTANT DISTINCTION

A public PortalDot receipt and matching TransferExecuted event are confirmed. The authorized amount was only 1 base unit (1e-18 EVM POT), so this proves the call path, not a meaningful-value payment.

Next steps and ecosystem value

A public testnet transfer is confirmed at 1 base unit; the next proof is a fresh authorization for a non-dust amount.

- **Open the live Web3 guard console**

  Inspect the current chain, contract code, guard balance and authority signer; connect a wallet in read-only mode.

- **Repeat with a non-dust amount**

  Use a fresh signed ALLOW; verify the intended recipient, receipt, TransferExecuted fields and balance delta.

- **Record demo and negative cases**

  Show live receipt evidence and prove denied, altered or expired authorizations are rejected.

ECOSYSTEM VALUE

A reusable spending-control component, a minimal guard contract, deployment evidence, fail-closed network tooling, an operations runbook and a live read-only Web3 console.

LIMITATIONS

Testnet prototype. No independent audit. EVM/SS58 mapping remains unverified. A non-dust transfer and the demo video remain pending.